

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250268241

Kind Code

A1

Publication Date

August 28, 2025

Inventor(s)

Flora; Joe et al.

SYSTEM FOR KEEPING LIVE INSECTS OR ANIMALS, AND METHOD

Abstract

A system for keeping live insects or animals. The system has a shelf assembly having at least a bottom wall and a rear wall. The bottom wall has opposite interior and exterior bottom wall surfaces and an outer perimeter with a rear face; the rear wall being mounted to extend upwardly from the rear face of the bottom wall. The rear wall having opposite interior and exterior rear wall surfaces. An airflow plenum is along the interior rear wall surface. The plenum has an airflow opening arrangement. A powered fan is in airflow communication with the plenum.

Inventors: Flora; Joe (Simi Valley, CA), Robbins; Chad (Middleton, WI)

Applicant: Spectrum Brands, Inc. (Middleton, WI)

Family ID: 1000008478761

Appl. No.: 19/061675

Filed: February 24, 2025

Related U.S. Application Data

us-provisional-application US 63558202 20240227

Publication Classification

Int. Cl.: A01K67/34 (20250101)

U.S. Cl.:

CPC A01K67/34 (20250101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. provisional patent application 63/558,202, filed on Feb. 27, 2024; the disclosure of which is incorporated herein by reference in its entirety.

TECHNICAL FIELD

[0002] This disclosure relates to systems and arrangements for keeping live insects or animals, such as in a retail setting. In particular, this disclosure relates to an arrangement having ventilation to aid in sustaining life of insects or animals in retail settings, and methods of their use.

BACKGROUND

[0003] Some people keep reptiles, such as snakes or lizards as pets. Birds, rats, and frogs are also kept as pets. Many reptiles, birds, frogs, and rats have live insects, such as crickets, as part of their diet. Retail locations specializing in pets and pet food can find it helpful to sell live insects, such as crickets, to customers who keep pets that would eat insects, such as crickets.

[0004] One problem is how to keep a large quantity of insects/crickets available for sale and in a convenient and easily accessible location in the retail setting, while also preventing fatality of them. Improvements are desirable.

SUMMARY

[0005] In general, a system for keeping live insects or animals is provided that improves the prior art.

[0006] In one aspect, a system for keeping live insects or animals is provided. The system comprises: a shelf assembly having at least a bottom wall and a rear wall; the bottom wall having opposite interior and exterior bottom wall surfaces and an outer perimeter with a rear face; the rear wall being mounted to extend upwardly from the rear face of the bottom wall; the rear wall having opposite interior and exterior rear wall surfaces; an airflow plenum along the interior rear wall surface; the plenum having an airflow opening arrangement; and a powered fan in airflow communication with the plenum.

[0007] In example embodiments, there further includes a plurality of support members holding the bottom wall away from a ground surface.

[0008] In some embodiments, the plenum is mounted against the interior bottom wall surface; the bottom wall includes a fan-holding region; the fan-holding region being within the plenum; and the powered fan is held within the fan-holding region.

[0009] In some arrangement, the support members are legs.

[0010] In examples, the support members may be casters.

[0011] In example embodiments, the outer perimeter of the bottom wall has a front face opposite of the rear face, and first and second side faces extending between the front face and rear face; the shelf assembly further includes a first side wall and a second side wall; the first side wall being along the first side face of the bottom wall; and the second side wall being along the second side face of the bottom wall.

[0012] In some arrangements, there further includes a slidable panel mounted to slide over the bottom wall and away from the rear wall.

[0013] Some embodiments further include a portable first bin; the first bin including an access opening and a vent opening; the first bin being positionable in the shelf assembly with the vent opening in communication with the airflow opening arrangement of the plenum.

[0014] In examples, the shelf assembly further includes a first shelf member mounted against the interior rear wall surface, and spaced above the interior bottom wall surface a sufficient distance to accommodate a bin between the first shelf member and bottom wall; and the shelf assembly defines a lower bin holding region between the first shelf member and bottom wall; and an upper bin

holding region above the first shelf member.

[0015] In some arrangements, the plenum extends along the interior rear wall surface to extend beyond the first shelf member into the upper bin holding region; and the airflow opening arrangement in the plenum includes at least a lower opening in the lower bin holding region, and an upper opening in the upper bin holding region.

[0016] Some embodiments further include a portable second bin; the second bin including an access opening and a vent opening; the second bin being positionable in the upper bin holding region with the vent opening of the second bin in communication with the upper opening of the plenum.

[0017] In some examples, the shelf system further includes a top wall extending at least partially over the upper bin holding region.

[0018] Example arrangements can further include a peg board back extending from the rear wall and above the top wall.

[0019] In some systems, the shelf assembly defines a double bin holding region along the bottom wall and between the first side wall and second side wall; the double bin holding region being sized to accommodate a portable first bin and a portable second bin.

[0020] In some embodiments, the airflow opening arrangement in the plenum includes at least a first opening and a second opening; the first opening and second opening being spaced laterally apart; the first opening positioned to be in communication with the portable first bin, and the second opening positioned to be in communication with the portable second bin.

[0021] Some example arrangement include the portable first bin including an access opening and a vent opening positioned to be in communication with the first airflow opening arrangement of the plenum; and the portable second bin including an access opening and a vent opening positioned to be in communication with the second airflow opening arrangement of the plenum.

[0022] The shelf system can further include a top wall extending at least partially over the double bin holding region.

[0023] The system can further include a peg board back extending from the rear wall and above the top wall.

[0024] Some embodiments further including a bag holder secured to the shelf system.

[0025] In some arrangements, the first bin includes a holding section defining an interior and a transparent lid moveably attached to the holding section; the first bin having a plurality of vent openings in the holding section.

[0026] In some embodiments, the powered fan moves between 30-50 cu. ft/minute.

[0027] In another aspect a method of keeping live insects or animals is provides. The method comprising: providing a shelf assembly having at least a bottom wall and a rear wall; the bottom wall having opposite interior and exterior bottom wall surfaces and an outer perimeter with a rear face; the rear wall being mounted to extend upwardly from the rear face of the bottom wall; the rear wall having opposite interior and exterior rear wall surfaces; providing an airflow plenum along the interior rear wall surface; the plenum having an airflow opening arrangement; and using a fan to provide air into the plenum.

[0028] The method can further include positioning a portable first bin having a vent opening in the shelf assembly with the vent opening in communication with the airflow opening arrangement of the plenum.

[0029] In some example methods, the shelf assembly further includes a first shelf member mounted against the interior rear wall surface, and spaced above the interior bottom wall surface a sufficient distance to accommodate a bin between the first shelf member and bottom wall; the shelf assembly defines a lower bin holding region between the first shelf member and bottom wall; and an upper bin holding region above the first shelf member; the plenum extends along the interior rear wall surface to extend beyond the first shelf member into the upper bin holding region; and the airflow opening arrangement in the plenum includes at least a lower opening in the lower bin holding

region, and an upper opening in the upper bin holding region.

[0030] The method may further include providing a portable second bin; the second bin including a vent opening; and positioning the second bin in the upper bin holding region with the vent opening of the second bin in communication with the upper opening of the plenum.

[0031] In some examples, the shelf assembly defines a double bin holding region along the bottom wall and between the first side wall and second side wall; and the method can include positioning a portable first bin and a portable second bin in the double bin holding region.

[0032] In some arrangements, the airflow opening arrangement in the plenum includes at least a first opening and a second opening; the first opening and second opening being spaced laterally apart; the first opening positioned to be in communication with the portable first bin, and the second opening positioned to be in communication with the portable second bin.

[0033] The method may further include putting insects in the portable first bin and portable second bin.

[0034] A variety of examples of desirable features or methods are set forth in part in the description that follows, and in part will be apparent from the description, or may be learned by practicing various aspects of the disclosure. The aspects of the disclosure may relate to individual features as well as combinations of features. It is to be understood that both the foregoing general description and the following detailed description are explanatory only, and are not restrictive of the claimed invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0035] FIG. 1 is a perspective view of a first embodiment of a shelf assembly system for keeping live insects or animals, constructed in accordance with principles of this disclosure;

[0036] FIG. 2 is a front view of the shelf assembly system of FIG. 1;

[0037] FIG. 3 is a side view of the shelf assembly system of FIG. 1;

[0038] FIG. 4 is a perspective view of a portable bin, used with the system of FIG. 1;

[0039] FIG. 5 is a partial, bottom, perspective view of the shelf assembly system of FIG. 1;

[0040] FIG. 6 is a perspective view of a second embodiment of a shelf assembly system for keeping live insects or animals, constructed in accordance with principles of this disclosure;

[0041] FIG. 7 is a front view of the shelf assembly system of FIG. 6;

[0042] FIG. 8 is a side view of the shelf assembly system of FIG. 6;

[0043] FIG. 9 is a partial, bottom, perspective view of the shelf assembly system of FIG. 6; and

[0044] FIG. 10 is a perspective view of the shelf assembly of FIG. 1, but with the bin removed from the lower bin holding region.

DETAILED DESCRIPTION

[0045] FIGS. 1 and 6 depict two example embodiments of a system for keeping live insects or animals; the system shown generally at 20. The system 20 of FIG. 1 is a stacked shelf system 22, while the system 20 of FIG. 6 is a side-by-side shelf system 24.

[0046] The system 20 includes a shelf assembly 26. By the term “shelf assembly”, it is meant a unit of components fitted, connected, fastened, or formed together having at least one or more rigid surfaces for the storage or display of one or more objects. The rigid surface is typically mostly flat and planar, but can have some variations.

[0047] The shelf assembly 26 includes a bottom wall 28. The bottom wall 28 is a rigid material, often sheet metal, or a hard plastic, or wood, which is located adjacent to or against a ground or floor. The bottom wall 28 has an interior facing wall surface 30, an opposite exterior facing wall surface 32, and an outer perimeter 34 surrounding the interior wall surface 30 and exterior wall surface 32.

[0048] In the example embodiment shown, the interior wall surface **30** and exterior wall surface **32** are generally flat, and planar. The outer perimeter **34** is rectangular in shape, but could be other shapes. In this example, the rectangular outer perimeter has a rear face **36**, opposite front face **38**, a first side face **40**, and a second side face. The first side face **40** and second side face extend between the front face **38** and rear face **36**.

[0049] The shelf assembly **26** further includes a rear wall **44**. The rear wall **44** is a rigid material, often sheet metal, or a hard plastic, or wood, which is mounted to extend upwardly from the rear face **36** of the bottom wall **28**. The rear wall **44** has an interior facing rear wall surface **46** and an opposite exterior facing rear wall surface **48**. In general, the interior rear wall surface **46** and exterior rear wall surface **48** are flat and planar. The rear wall **44** is rectangular in shape, but could be other shapes.

[0050] The system **20** further includes an airflow channel or plenum **50**. The plenum **50** is along the interior rear wall surface **46** and includes an airflow opening arrangement **52**.

[0051] In one example embodiment, the plenum **50** is mounted against the interior bottom wall surface **30**. The plenum **50** is constructed and arranged to allow for the flow of air therethrough.

[0052] The system **20** further includes a powered fan **54**. The fan **54** is positioned to be in airflow communication with the plenum **50** to draw the flow of air through the plenum **50**. In one example, the powered fan moves between 30-50 cu. ft/minute, with enough power to circulate air, but with not so much power that it disturbs the insects/animals held in the system **20**.

[0053] In example arrangements, the bottom wall **28** includes a fan-holding region **56**. The fan-holding region **56** is located within the interior of the plenum **50** and is sized to hold the fan **54**. The fan **54** is operably held within the fan-holding region **56**. More details of the plenum **50** and fan **54** are discussed further below, in connection with the embodiment of FIGS. 1-5 and the embodiment of FIGS. 6-9.

[0054] The system **20** further includes a plurality of support members **58**. The support members **58** hold the bottom wall **28** away from a ground surface. The support members **58** can be legs **60**, including short legs **60** (FIG. 1) or longer legs **60** (FIG. 6). In other arrangements, the support members **58** can be casters.

[0055] The shelf assembly **26** further includes a first side wall **62**. The first side wall **62** has an interior-facing surface **64**, and an opposite exterior-facing surface **66**. The first side wall **62** has a perimeter edge **68**. The perimeter edge **68** can form many different shapes, and in the examples shown, the perimeter edge **68** is trapezoidal in shape. The perimeter edge **68** has a bottom edge **70**, which opposes or faces the ground surface; an opposite top edge **72**; a rear edge **74**; an opposite front edge **76**; and an angled edge **78** extending between the front edge **76** and top edge **72**. The first side wall **62** a rigid material, often sheet metal, or a hard plastic, or wood.

[0056] The shelf assembly **26** further includes a second side wall **80**. The second side wall **80** has an interior-facing surface **82**, and an opposite exterior-facing surface **84**. The second side wall **80** has a perimeter edge **86**. The perimeter edge **86** can form many different shapes, and in the examples shown, the perimeter edge **86** is trapezoidal in shape. The perimeter edge **86** has a bottom edge **88**, which opposes or faces the ground surface; an opposite top edge **90**; a rear edge; an opposite front edge **94**; and an angled edge **96** extending between the front edge **94** and top edge **90**. The second side wall **80** a rigid material, often sheet metal, or a hard plastic, or wood.

[0057] The first side wall **62** and the second side wall **80** are positioned to be against opposite faces of the bottom wall **28**, such that the interior facing surfaces **64**, **82** are opposing each other. That is, the first side wall **62** is along the first side face **40** of the bottom wall **28**, while the second side wall **80** is along the second side face **42** of the bottom wall **28**.

[0058] The shelf assembly **26** further includes a top wall **140**. The top wall **140** has an inner facing surface **142** and an opposite exterior facing surface **144**. In the example embodiment, the top wall **140** has a rectangular perimeter edge including a front edge **146**, rear edge **148**, and first and second side edges **150**, **152**. The top wall **140** extends horizontally between the first side wall **62**

and second side wall **80**. The inner facing surface **142** faces toward the remainder of the shelf assembly **26**. The first and second side edges **150**, **152** are adjacent to the top edges **72**, **90** of the first side wall **62** and second side wall **80**.

[0059] The system **20** can further include an optional pegboard back **154**. The pegboard back **154** extends from the rear wall **44** and above the top wall **140**. The pegboard back **154** is useable for holding useful items, such as rubber band containers **156**, calcium powder containers **158**, a holder for a cricket counter **160**, etc. A section **162** can also be used to hold signage.

[0060] The system **20** further includes a portable bin for holding live insects or animals. In the example embodiments shown, there is a portable first bin **100** and a portable second bin **102**. The system **20** can be designed to accommodate more bins, as well. Both bins **100**, **102** have the same features, so in describing the example embodiment shown in FIG. **4**, it should be understood that both bins **100**, **102** are being described.

[0061] The bins **100**, **102** have a base **104**, a surrounding wall **106** extending up from the base **104**, and a cover **108** secured on top of the surrounding wall **106**. The surrounding wall **106** includes a rear wall **110**, first and second side walls **112**, **114**, and a front wall **116**. The front wall **116** angles outwardly from the base **104** at a non-perpendicular angle to project away from the base **104**. The base **104**, surrounding wall **106**, and cover **108** enclose an interior **118**, which functions as a holding section **120** for holding crickets, insects, or other live animals.

[0062] The cover **108** includes a transparent lid **122**. The lid **122** allows for viewing into the holding section **120**. The lid **122** is moveably attached to the holding section **120**, and in the example shown, is pivotably attached to a remainder of the cover **108**. The lid **122** pivotably covers an access opening **124**, providing access to the interior **118**. A customer can view the interior **118** of the bin **100**, **102** through the lid **122**, and then pivot the lid **122** to expose the access opening **124**. When the access opening **124** is exposed, the customer can reach into the interior **118** and remove crickets, insects, or other live animals being kept in the holding section **120** of the bins **100**, **102**.

[0063] The bins **100**, **102** each has at least one vent opening **126**. The at least one vent opening **126** is located such that, when the bins **100**, **102** are positioned on the shelf assembly **26**, the vent opening **126** is put into airflow communication with the airflow opening arrangement **52** of the plenum **50**.

[0064] In the example embodiment illustrated, the vent opening **126** is in the rear wall **110** of the bin **100**, **102**. This way, the bin **100**, **102** can be positioned on the shelf assembly **26** with its rear wall **110** against the shelf assembly rear wall **44**, and with the transparent lid **122** facing outward from the shelf assembly **26**.

[0065] In preferred embodiments, each of the bins **100**, **102** has a plurality of vent openings **128**, **129**, **130**. The vent openings **128**, **129**, **130** are in the holding section **120**. In the example shown, the rear wall **110** has vent opening **126** and vent opening **128**, spaced laterally apart from each other, and centered between the top and bottom of the bin **100**, **102**. The cover **108** has vent openings **129**, **130** in a portion of the cover **108** that is spaced over and generally parallel to the base **104**. There can be many variations, and include more or fewer vent openings.

[0066] The vent openings **126**, **128**, **129**, **130** are depicted as round, but could be other shapes. The openings **126**, **128**, **129**, **130** are covered with a screen **131**, **132**, **133**, **134**.

[0067] The bins **100**, **102** can be made from plastic or other materials, such as metal or wood. FIG. **4** allows for viewing the interior **118**, but it should be understood that the surrounding wall **106** is typically opaque and not transparent. In the interior **118**, optional egg carton holders **136** are visible, resting against the base **104**. The egg carton holders **136** are used to hold egg cartons (not shown), which make for hospitable homes for crickets (or other insects) and provide for extra space in the interior **118**.

[0068] An optional bag holder **164** is secured to the shelf assembly **26**. In the example shown, the bag holder **164** is attached to the exterior facing surface **66** of the first side wall **62** to project therefrom. The bag holder **164** holds bags for packaging the crickets, insects, or other animals held

in the bins **100, 102**.

Stacked Shelf System, FIGS. 1-3

[0069] In reference now to the stacked shelf system **22** of FIGS. 1-3, the system **22** further includes a slidable panel **170**. The panel **170** is movably mounted to slide over the bottom wall **28** and toward/away from the rear wall **44**. The panel **170** extends horizontally between the first side wall **62** and second side wall **80**. The panel **170** includes a hand hold **172** at a free end along a front edge, which allows for convenient grasping by a human hand. The panel **170** is sized to hold one of the bins **100, 102**. In use, a person can grasp the hand hold **172** to pull the panel **170** out from a remaining part of the shelf system **22** to allow for better access to the bin **100, 102** resting on the panel **170**.

[0070] The shelf assembly **26** further includes a first shelf member **174**. In the example illustrated, the first shelf member **174** is mounted against the interior rear wall surface **46** and spaced above the interior bottom wall surface **30** a sufficient distance to accommodate a bin between the first shelf member **174** and the bottom wall **28**. The first shelf member **174** extends horizontally between the first side wall **62** and the second side wall **80**.

[0071] The shelf assembly **26** defines a lower bin holding region **176** between the first shelf member **174** and bottom wall **28**. In particular, the lower bin holding region **176** is between the first shelf member **174** and the slidable panel **170**. The lower bin holding region **176** is sized to accommodate one of the bins **100, 102**.

[0072] The shelf assembly **26** defines an upper bin holding region **178** above the first shelf member **174**. The top wall **140** extends at least partially over the upper bin holding region **178**. The upper bin holding region **178** is sized to accommodate one of the bins **100, 102**.

[0073] In reference now to FIG. 3, the plenum **50** extends along the interior rear wall surface **46** to extend beyond the first shelf member **174** into the upper bin holding region **178**. The airflow opening arrangement **52** in the plenum **50** includes at least a lower opening **180** in the lower bin holding region **176**, and an upper opening **182** in the upper bin holding region **178**.

[0074] The portable first bin **100** is positionable in the lower bin holding region **176**, such that the vent opening **126** can be against and in direct airflow communication with the lower opening **180** of the plenum **50**. The portable second bin **102** is positionable in the upper bin holding region **178**, such that the vent opening **126** of the second bin **102** can be against and in direct airflow communication with the upper opening **182** of plenum **50**.

[0075] The system **20** includes an aid **199** to align the vents **126** in the bins **100, 102** with the openings **180, 182** in the plenum **50**. In FIG. 10, the aid is embodied as a plurality of projecting tabs **200**, arranged to project from an upper face **202** of the slidable panel **170** and from an upper face **204** of the first shelf member **174**. The tabs **200** are located such that an outside perimeter of the base **104** of the bins **100, 102** is closely contained within the tabs **200**. The tabs **200** can be small, sheet metal tabs projections.

[0076] In reference now to FIG. 3, when the fan **54** is operating, air is drawn through the bins **100, 102**, and from the bins through vent opening **126**, then through the respective lower opening **180** or upper opening **182**, through the plenum **50** and out through the airflow opening arrangement **52**.

Side by Side Shelf System, FIGS. 6-9

[0077] In reference now to FIGS. 6-9, the side by side shelf system **24** defines a double bin holding region **190** along the bottom wall **28** and between the first side wall **62** and second side wall **80**.

The top wall **140** extends at least partially over the double bin holding region **190**. The double bin holding region **190** is sized to accommodate the portable first bin **100** and portable second bin **102**.

[0078] Attention is directed to FIG. 9. The airflow opening arrangement **52** in the plenum **50** includes at least a first opening **192** and a second opening **194**. The first opening **192** and second opening **194** are spaced laterally apart. The first opening **192** is positioned to be in communication with the portable first bin **100**, and the second opening **194** is positioned to be in communication with the portable second bin **102**.

[0079] The side by side shelf system **24** includes an aid to the align the vents **126** in the bins **100**, **102** with the openings **192**, **194** in the plenum **50**. The aid **199** is embodied as projecting tabs **200**, arranged to project from an upper face of the bottom wall **28**, analogous to the tabs **200** shown in FIG. **10**, projecting from the slidable panel **170** and from the first shelf member **174**. In the side by side shelf system **24**, there are two sets of tabs **200**. A first of the two sets of tabs **200** projects from the bottom wall **28** to outline where the portable first bin **100** is positioned, and a second of the two sets of tabs **200** projects from the bottom wall **28** to outline where the portable second bin **102** is positioned.

[0080] For example, the portable first bin **100** is positioned in the double bin holding region **190** such that the vent opening **126** is against the first opening **192** of the plenum **50** to be in airflow communication therewith. The portable second bin **102** is positioned in the double bin holding region **190** such that the vent opening **126** of the second bin **102** is against the second opening **194** of the plenum **50** to be in airflow communication therewith.

[0081] In reference to FIG. **8**, when the fan **54** is operating, air is drawn through the bins **100**, **102**, and from the bins through vent opening **126**, then through the respective first opening **192** or second opening **194**, through the plenum **50** and out through the airflow opening arrangement **52**.
Example Methods of Use

[0082] The above system **20** is usable in a method of keeping live insects or animals. The method includes providing the shelf assembly **26** having at least bottom wall **28** and rear wall **44**. The bottom wall **28** has opposite interior and exterior bottom wall surfaces **30**, **32** and outer perimeter **34** with rear face **36**. The rear wall **44** is mounted to extend upwardly from the rear face **36** of the bottom wall **28**. The rear wall **44** has opposite interior and exterior rear wall surfaces **46**, **48**.

[0083] The method includes providing airflow plenum **50** along the interior rear wall surface **46**. The plenum **50** has airflow opening arrangement **52**.

[0084] The method includes using fan **54** to provide air into the plenum **50**.

[0085] The method further includes positioning portable first bin **100** having vent opening **126** in the shelf assembly **26** with the vent opening **126** in communication with the airflow opening arrangement **52** of the plenum **50**.

[0086] The method further includes providing portable second bin **102** having vent opening **126** and positioning the second bin **102** in the upper bin holding region **178** with the vent opening **126** of the second bin **102** in communication with the upper opening of the plenum **182**.

[0087] In one example method, the shelf assembly **26** defines double bin holding region **190** along the bottom wall **28** and between the first side wall **62** and second side wall **80**. The method includes positioning portable first bin **100** and portable second bin **102** in the double bin holding region **190**. The airflow opening arrangement **52** in the plenum **50** includes at least first opening **192** and second opening **194**. The first opening **192** and second opening **194** are spaced laterally apart. The first opening **192** is positioned to be in communication with the portable first bin **100**, and the second opening **194** is positioned to be in communication with the portable second bin **102**.

[0088] The method can further including putting insects in the portable first bin **100** and portable second bin **102**. The method can further include removing insects from the first bin **100** or second bin **102**. These steps can include pivoting the lid **122** to expose the access opening **124** into the interior **118** of the bin **100**, **102**.

[0089] The method can further include grasping hand hold **172**, sliding the panel **170** out and away from the rear wall **44** and then accessing the bin **100**, **102** in the lower bin holding region **176**.

[0090] The above represents example principles. Many embodiments can be made using these principles.

Claims

- 1.** A system for keeping live insects or animals; the system comprising: (a) a shelf assembly having at least a bottom wall and a rear wall; (i) the bottom wall having opposite interior and exterior bottom wall surfaces and an outer perimeter with a rear face; (ii) the rear wall being mounted to extend upwardly from the rear face of the bottom wall; the rear wall having opposite interior and exterior rear wall surfaces; (b) an airflow plenum along the interior rear wall surface; the plenum having an airflow opening arrangement; and (c) a powered fan in airflow communication with the plenum.
- 2.** The system of claim 1 further including a plurality of support members holding the bottom wall away from a ground surface.
- 3.** The system of claim 2, wherein: (a) the plenum is mounted against the interior bottom wall surface; (b) the bottom wall includes a fan-holding region; (i) the fan-holding region being within the plenum; and (c) the powered fan is held within the fan-holding region.
- 4.** The system of claim 2, wherein the support members are legs.
- 5.** The system of claim 2, wherein the support members are casters.
- 6.** The system of claim 1, wherein: (a) the outer perimeter of the bottom wall having a front face opposite of the rear face, and first and second side faces extending between the front face and rear face; (b) the shelf assembly further includes a first side wall and a second side wall; (i) the first side wall being along the first side face of the bottom wall; and (ii) the second side wall being along the second side face of the bottom wall.
- 7.** The system of claim 6 further including a slidable panel mounted to slide over the bottom wall and away from the rear wall.
- 8.** The system of claim 1, further including a portable first bin; the first bin including an access opening and a vent opening; the first bin being positionable in the shelf assembly with the vent opening in communication with the airflow opening arrangement of the plenum.
- 9.** The system of claim 1, wherein: (a) the shelf assembly further includes a first shelf member mounted against the interior rear wall surface, and spaced above the interior bottom wall surface a sufficient distance to accommodate a bin between the first shelf member and bottom wall; and (b) the shelf assembly defines a lower bin holding region between the first shelf member and bottom wall; and an upper bin holding region above the first shelf member.
- 10.** The system of claim 9 wherein: (a) the plenum extends along the interior rear wall surface to extend beyond the first shelf member into the upper bin holding region; and (b) the airflow opening arrangement in the plenum includes at least a lower opening in the lower bin holding region, and an upper opening in the upper bin holding region.
- 11.** The system of claim 8, further including a portable second bin; the second bin including an access opening and a vent opening; the second bin being positionable in the upper bin holding region with the vent opening of the second bin in communication with the upper opening of the plenum.
- 12.** The system of claim 9, wherein the shelf system further includes a top wall extending at least partially over the upper bin holding region.
- 13.** The system of claim 12, further including a peg board back extending from the rear wall and above the top wall.
- 14.** The system of claim 6, wherein the shelf assembly defines a double bin holding region along the bottom wall and between the first side wall and second side wall; the double bin holding region being sized to accommodate a portable first bin and a portable second bin.
- 15.** The system of claim 14 wherein the airflow opening arrangement in the plenum includes at least a first opening and a second opening; the first opening and second opening being spaced laterally apart; the first opening positioned to be in communication with the portable first bin, and the second opening positioned to be in communication with the portable second bin.
- 16.** The system of claim 15 further including: (a) the portable first bin including an access opening

and a vent opening positioned to be in communication with the first airflow opening arrangement of the plenum; and (b) the portable second bin including an access opening and a vent opening positioned to be in communication with the second airflow opening arrangement of the plenum.

17. The system of claim 14, wherein the shelf system further includes a top wall extending at least partially over the double bin holding region.

18. The system of claim 17, further including a peg board back extending from the rear wall and above the top wall.

19. The system of claim 1, further including a bag holder secured to the shelf system.

20. The system of claim 8, wherein the first bin includes a holding section defining an interior and a transparent lid moveably attached to the holding section; the first bin having a plurality of vent openings in the holding section.

21. The system of claim 8, wherein the shelf assembly includes projecting tabs to aid in positioning the first bin with the vent opening in communication with the airflow opening arrangement of the plenum.

22. The system of claim 1, wherein the powered fan moves between 30-50 cu. ft/minute.

23. A method of keeping live insects or animals; the method comprising: (a) providing a shelf assembly having at least a bottom wall and a rear wall; (i) the bottom wall having opposite interior and exterior bottom wall surfaces and an outer perimeter with a rear face; (ii) the rear wall being mounted to extend upwardly from the rear face of the bottom wall; the rear wall having opposite interior and exterior rear wall surfaces; (b) providing an airflow plenum along the interior rear wall surface; the plenum having an airflow opening arrangement; and (c) using a fan to provide air into the plenum.

24. The method of claim 23 further including: positioning a portable first bin having a vent opening in the shelf assembly with the vent opening in communication with the airflow opening arrangement of the plenum.

25. The method of claim 24, wherein a step of positioning the portable first bin includes positioning the portable first bin within a set of projecting tabs in the shelf assembly.

26. The method of claim 24, wherein: (a) the shelf assembly further includes a first shelf member mounted against the interior rear wall surface, and spaced above the interior bottom wall surface a sufficient distance to accommodate a bin between the first shelf member and bottom wall; (b) the shelf assembly defines a lower bin holding region between the first shelf member and bottom wall; and an upper bin holding region above the first shelf member; (c) the plenum extends along the interior rear wall surface to extend beyond the first shelf member into the upper bin holding region; and (d) the airflow opening arrangement in the plenum includes at least a lower opening in the lower bin holding region, and an upper opening in the upper bin holding region.

27. The method of claim 26 further including providing a portable second bin; the second bin including a vent opening; and positioning the second bin in the upper bin holding region with the vent opening of the second bin in communication with the upper opening of the plenum.

28. The method of claim 24, wherein the shelf assembly defines a double bin holding region along the bottom wall and between the first side wall and second side wall; and positioning a portable first bin and a portable second bin in the double bin holding region.

29. The method of claim 28, wherein the airflow opening arrangement in the plenum includes at least a first opening and a second opening; the first opening and second opening being spaced laterally apart; the first opening positioned to be in communication with the portable first bin, and the second opening positioned to be in communication with the portable second bin.

30. The method of claim 27, further including putting insects in the portable first bin and portable second bin.
